

SAP BTP Integration Development

Employee Data Synchronization (SF to S/4HANA)

using SAP Cloud Integration

CAPSTONE PROJECT

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1. Project Overview

Introduction

The project implements an integration solution using SAP Integration Suite – Cloud Integration to transfer employee data from SAP SuccessFactors to SAP S/4HANA.

The integration flow retrieves employee information from SuccessFactors, processes and maps the data into the required S/4HANA structure, and sends the processed data to the target system.

The solution also includes SFTP file processing and email notifications for successful and failed processing scenarios.

Project Objective

The main objectives of the project are:

- Retrieve employee data from SAP SuccessFactors.
- Transform the source data according to the target structure.
- Send the processed data to SAP S/4HANA.
- Convert the processed XML data into CSV format.
- Upload the CSV file to an SFTP server.
- Send email notifications for successful processing.
- Handle runtime errors using an Exception Subprocess.
- Send failure notifications through email.

Technologies Used

Technology / Component	Purpose
SAP Cloud Integration	Design and execution of iFlows
SAP SuccessFactors	Source system
SAP S/4HANA	Target system
SFTP Adapter	File transfer
Files.com	Cloud storage
Mail Adapter	Email notification
Message Mapping	Data transformation
XML to CSV Converter	Format conversion
Exception Subprocess	Error handling
Timer Start Event	Scheduled execution
Request Reply	Sends a request to a receiver and waits for the response

2. Business Requirement

The requirement is to automate the transfer of employee information from SAP SuccessFactors to SAP S/4HANA.

Instead of manually extracting and transferring employee information, the integration flow performs the required processing automatically.

The integration should also provide a mechanism to store processed information as a CSV file on an SFTP server and notify the responsible user through email when processing is successful or when an error occurs.

3. Solution Overview

The integration solution consists of a scheduled integration flow that communicates with multiple systems.

The flow starts with a Timer Start Event. It retrieves employee data from SuccessFactors using the SuccessFactors adapter. The retrieved data is then transformed using Message Mapping and sent to S/4HANA through an OData-based Request Reply process.

After receiving the response, the flow uses a Sequential Multicast to perform additional processing. One branch converts the XML data into CSV and uploads it to the SFTP server. Another branch prepares an email notification.

An Exception Subprocess is also implemented to handle runtime errors and send a failure notification through email.

4. Integration Architecture

Main Integration Flow

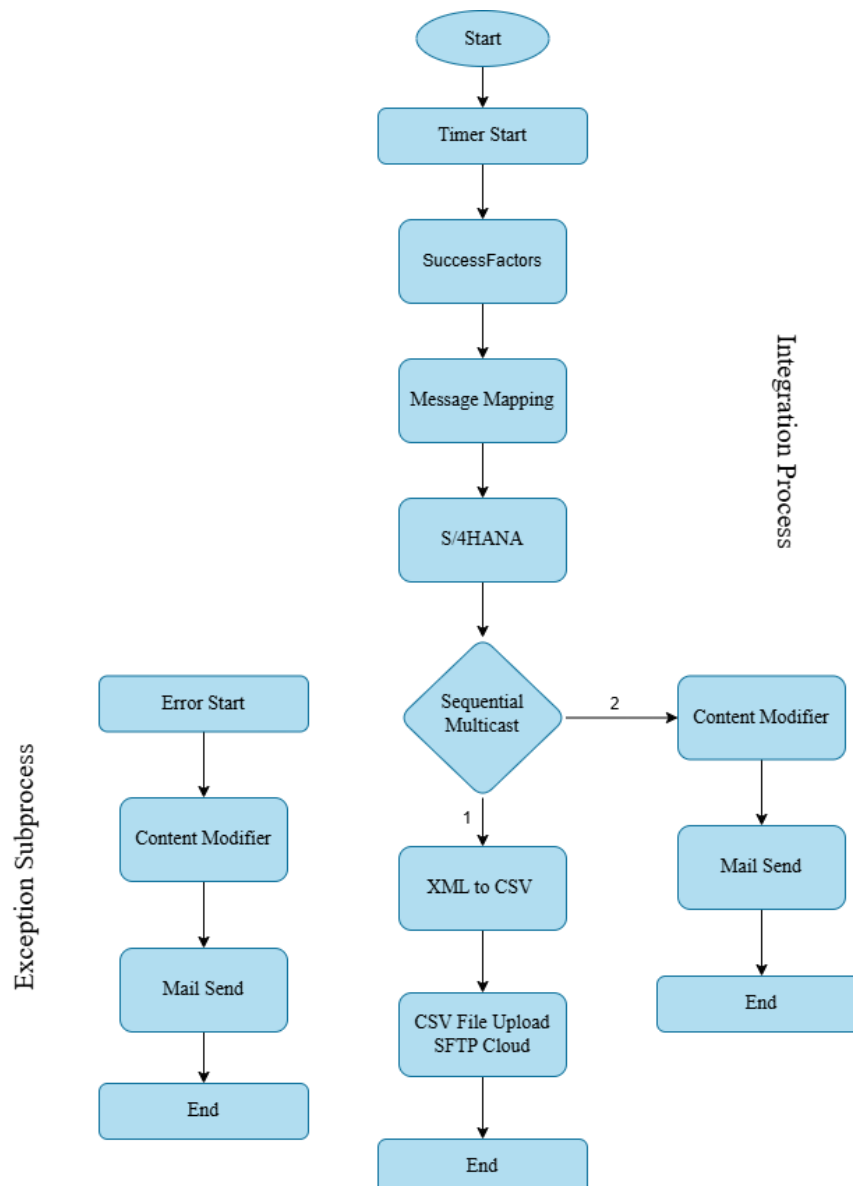


Figure 1: Overall Integration Flowchart Architecture

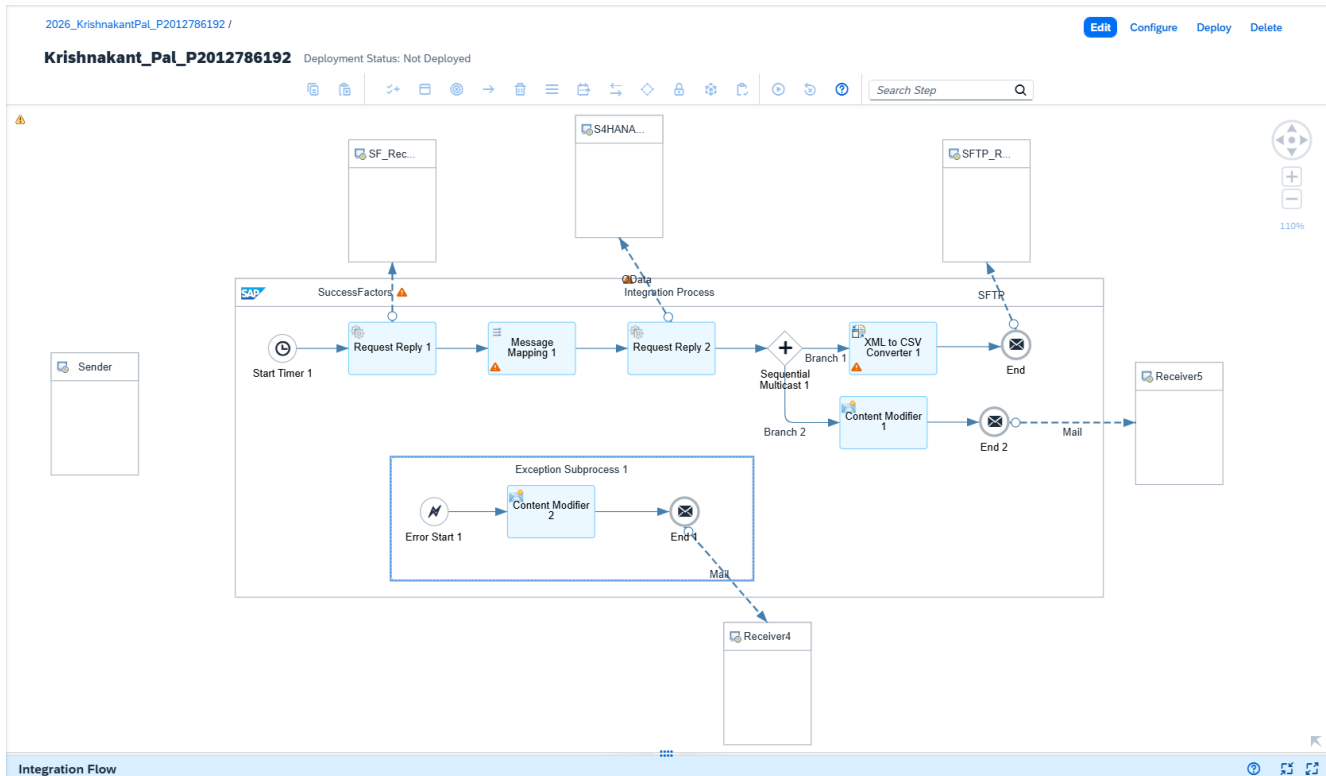


Figure 2: Overall SAP SuccessFactors to S/4HANA Integration Flow

5. Integration Flow Components

The following components are used in the integration flow.

Component	Description
Timer Start	Starts the integration flow according to the configured schedule.
Request Reply	Sends a request to the configured receiver and waits for its response.
SuccessFactors Adapter	Retrieves employee data from SAP SuccessFactors.
Message Mapping	Maps source employee fields to the target S/4HANA structure.
OData Adapter (V2)	Communicates with the S/4HANA OData service.
Sequential Multicast	Splits the processing into sequential branches.
XML to CSV Converter	Converts XML data into CSV format.
SFTP Adapter	Uploads the generated CSV file to the SFTP server.
Content Modifier	Creates or modifies message content used for notification.
Mail Adapter	Sends email notifications.
Exception Subprocess	Handles errors occurring during integration processing.

6. SAP SuccessFactors Configuration

SuccessFactors Adapter

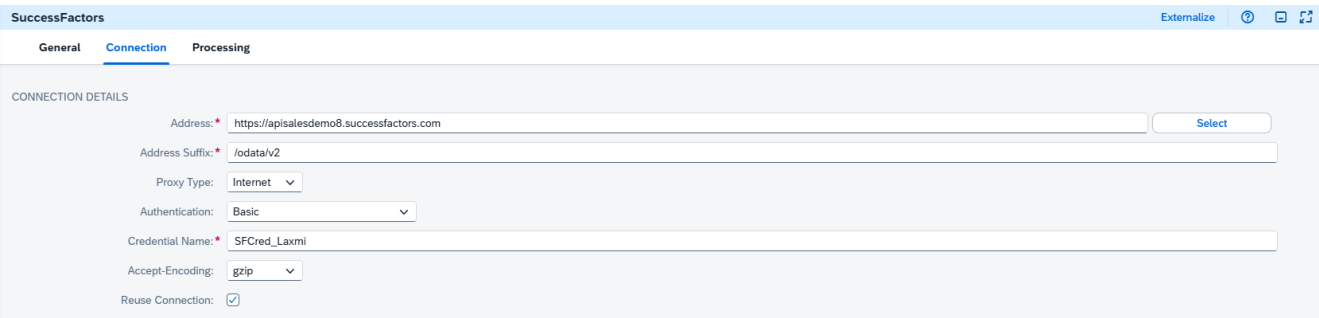
The SuccessFactors adapter is used to retrieve employee information from the SAP SuccessFactors system.

The adapter is configured with the required connection details and uses the User resource to retrieve employee data.

Operation: Query (GET)

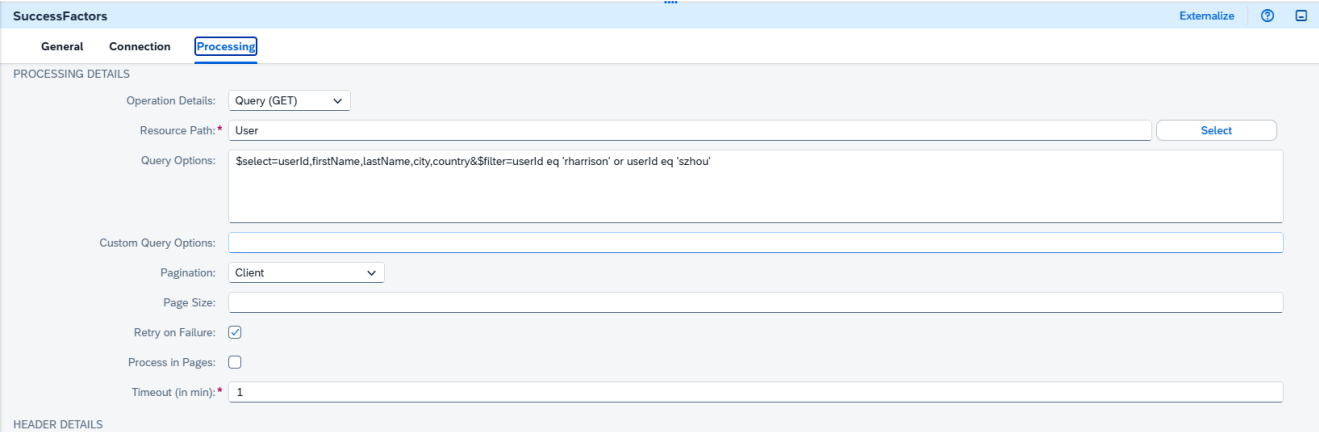
Resource Path: User

Query Options: \$select=userid,firstName,lastName,city,country&\$filter=userid eq 'harrison' or userid eq 'szhou'



The screenshot shows the 'Connection' tab of the SAP SuccessFactors Adapter Configuration. It includes fields for Address (https://apisalesdemo8.successfactors.com), Address Suffix (/odata/v2), Proxy Type (Internet), Authentication (Basic), Credential Name (SFCred_Laxmi), Accept-Encoding (gzip), and a checked Reuse Connection checkbox.

Figure 3: SAP SuccessFactors Adapter Configuration (Connection Tab)



The screenshot shows the 'Processing' tab of the SAP SuccessFactors Adapter Configuration. It includes fields for Operation Details (Query (GET)), Resource Path (User), Query Options (\$select=userid,firstName,lastName,city,country&\$filter=userid eq 'harrison' or userid eq 'szhou'), Custom Query Options, Pagination (Client), Page Size, Retry on Failure (checked), Process in Pages (unchecked), and Timeout (1 min).

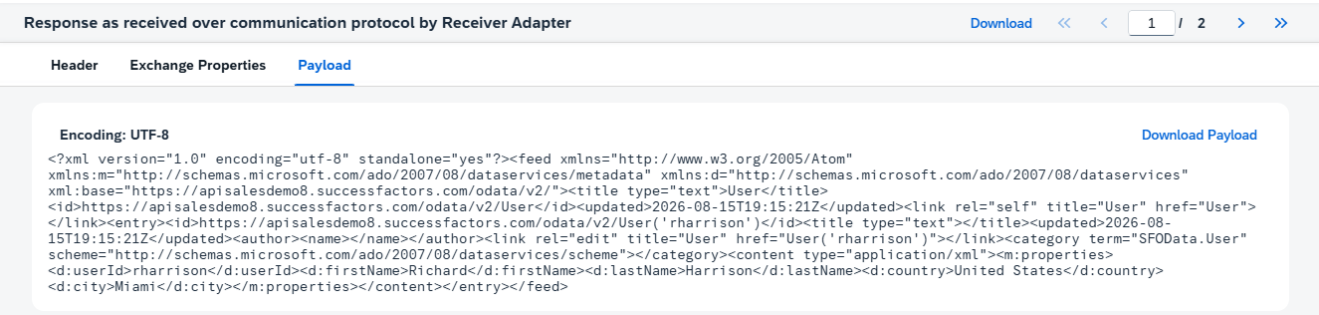
Figure 4: SAP SuccessFactors Adapter Configuration (Processing Tab)

SuccessFactors Data Retrieval

The integration flow retrieves the selected employee fields from SuccessFactors.

The following fields are retrieved:

Field	Description
userid	Employee/User identifier
firstName	Employee first name
lastName	Employee last name
city	Employee city
country	Employee country



The screenshot shows the 'Payload' tab of the SAP SuccessFactors Adapter Configuration. It displays the XML response received over communication protocol by Receiver Adapter. The XML content is as follows:

```
<?xml version="1.0" encoding="utf-8" standalone="yes"?><feed xmlns="http://www.w3.org/2005/Atom"
xmlns:m="http://schemas.microsoft.com/ado/2007/08/dataservices/metadata" xmlns:d="http://schemas.microsoft.com/ado/2007/08/dataservices"
xml:base="https://apisalesdemo8.successfactors.com/odata/v2/"><title type="text">User</title>
<id>https://apisalesdemo8.successfactors.com/odata/v2/User</id><updated>2026-08-15T19:15:21Z</updated><link rel="self" title="User" href="User">
</link><entry><id>https://apisalesdemo8.successfactors.com/odata/v2/User('rharrison')</id><title type="text"></title><updated>2026-08-
15T19:15:21Z</updated><author><name></name></author><link rel="edit" title="User" href="User('rharrison')"></link><category term="SF0Data.User"
scheme="http://schemas.microsoft.com/ado/2007/08/dataservices/scheme"></category><content type="application/xml"><m:properties>
<d:userId>rharrison</d:userId><d:firstName>Richard</d:firstName><d:lastName>Harrison</d:lastName><d:country>United States</d:country>
<d:city>Miami</d:city></m:properties></content></entry></feed>
```

Figure 5: Employee Data Retrieved from SuccessFactors

7. Message Mapping

Mapping Overview

Message Mapping is used to transform the employee data received from SuccessFactors into the structure required by the S/4HANA target system.

The source structure contains the User information, while the target structure contains EmpdetailsSet and Empdetails.

Source and Target Structure

The source fields include	The target structure contains
User └─ User ├─ userId ├─ firstName ├─ lastName ├─ city └─ country	EmpdetailsSet └─ Empdetails ├─ Mandt ├─ Participantid ├─ Sfempname ├─ City └─ Country

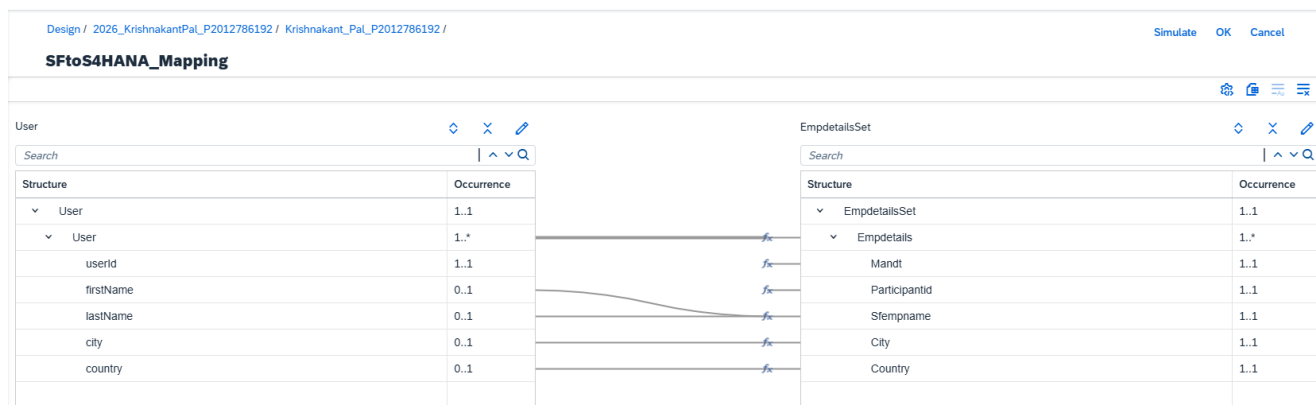


Figure 6: SuccessFactors to S/4HANA Message Mapping

The required source fields are mapped to their corresponding target fields.

Participant ID Mapping

The Participantid field is populated according to the mapping configuration.

During testing, it was also observed that using an already existing Participant ID can result in a duplicate-record error from the target S/4HANA system. This scenario was used as one of the negative test cases.

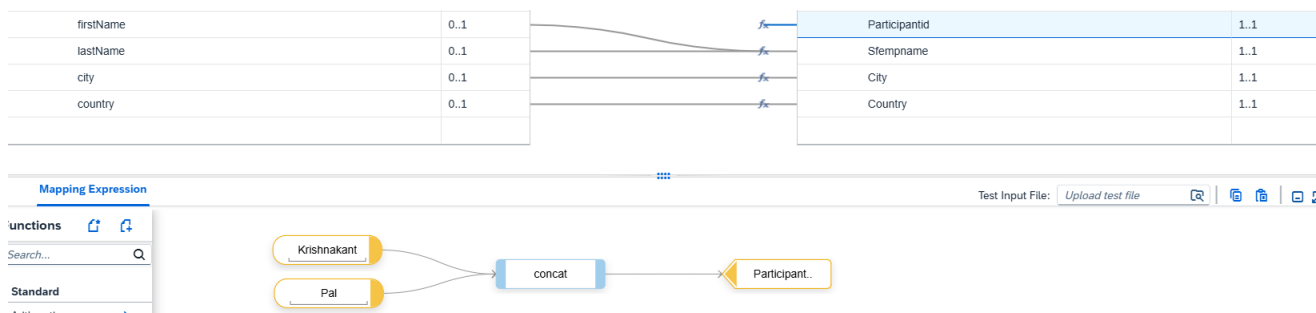


Figure 7: Participant ID Message Mapping Expression

8. S/4HANA Configuration

S/4HANA Integration

The transformed employee data is sent to the SAP S/4HANA system using the configured OData service.

The integration uses a Request Reply step to send the request and receive the response from S/4HANA.

The target system validates the incoming employee information and returns the processing response.

The screenshot shows the 'Connection' tab of the SAP S/4HANA OData Receiver Configuration. The 'CONNECTION DETAILS' section includes the following fields: Address (http://awslkm0133.training.accenture.com:8000/sap/opu/odata/sap/ZEMPDET_SRV_SRV), Proxy Type (On-Premise), Location ID, Authentication (Basic), Credential Name (L4C_Cred), CSRF Protected (checked), and Reuse Connection (checked).

Figure 8: SAP S/4HANA OData Receiver Configuration (Connection Tab)

The screenshot shows the 'Processing' tab of the SAP S/4HANA OData Receiver Configuration. The 'PROCESSING DETAILS' section includes the following fields: Operation Details (Create (POST)), Resource Path (EmpdetailsSet), Fields (Mandt, Participantid, Sfempname, City, Country), Enable Batch Processing (unchecked), Custom Query Options, Content Type (Atom), Content Type Encoding (None), Attach Error Details on Failure (checked), and Timeout (in min) (1). The 'HEADER DETAILS' section includes Request Headers (traceparent) and Response Headers (traceparent).

Figure 9: SAP S/4HANA OData Receiver Configuration (Processing Tab)

S/4HANA Response

After the request is processed, S/4HANA returns an OData response.

A successful response contains the processed employee information and an HTTP success status.

During testing, successful processing was verified through the Cloud Integration Message Monitor.

The screenshot shows the 'Payload' tab of the SAP S/4HANA OData Receiver Configuration. The 'Response as received over communication protocol by Receiver Adapter' section includes the following fields: Encoding (UTF-8), Download Payload, and the XML response body. The XML response body is as follows: <?xml version='1.0' encoding='utf-8'?><entry xml:base='http://awslkm0133.training.accenture.com:8000/sap/opu/odata/sap/ZEMPDET_SRV_SRV/' xmlns='http://www.w3.org/2005/Atom' xmlns:m='http://schemas.microsoft.com/ado/2007/08/dataservices/metadata' xmlns:d='http://schemas.microsoft.com/ado/2007/08/dataservices'><id>http://awslkm0133.training.accenture.com:8000/sap/opu/odata/sap/ZEMPDET_SRV_SRV/EmpdetailsSet(Mandt='500',Participantid='Krishnakant%20Pal')</id><title type='text'>EmpdetailsSet(Mandt='500',Participantid='Krishnakant%20Pal')</title><updated>2026-08-14T10:38:57Z</updated><category term='ZEMPDET_SRV_SRV.Empdetails' scheme='http://schemas.microsoft.com/ado/2007/08/dataservices/scheme'><link href='EmpdetailsSet(Mandt='500',Participantid='Krishnakant%20Pal')' rel='edit' title='Empdetails'></link><content type='application/xml'><m:properties><d:Mandt>500</d:Mandt><d:Sfempname>Richard Harrison</d:Sfempname><d:City>Miami</d:City><d:Country>United States</d:Country><d:Createdon>0000-00-00</d:Createdon></m:properties></content></entry>

Figure 10: Successful S/4HANA Response

9. XML to CSV Conversion

After receiving the S/4HANA response, the integration flow processes the XML data through the XML to CSV Converter.

The converter transforms the XML employee data into CSV format so that it can be stored on the configured SFTP server.

The screenshot shows the 'XML To CSV Converter' configuration window. It has two tabs: 'General' and 'Processing'. The 'Processing' tab is active. Under 'BASIC ELEMENTS', the 'Path to Source Element in XML' is set to '/EmpdetailsSet/Empdetails'. The 'Field Separator in CSV' is set to 'Comma(,)' and the 'CSV Header' is set to 'XML based'.

Figure 11: XML to CSV Conversion Configuration

The generated CSV contains employee-related fields such as:

Mandt,Participantid,Country,Sfempname,City,Createdon

The screenshot shows the 'Message Processing Run' interface. On the left, a 'Run Steps (17)' list shows the flow: SFTP (6 sec 185 ms), End (7 ms), XML to CSV Converter 1 (23 ms), and Sequential Multicast 1 (6 sec 215 ms). The 'XML to CSV Converter 1' step is highlighted. On the right, the 'Message before Step' tab shows the XML payload: <EmpdetailsSet><Empdetails><Mandt>500</Mandt><Participantid>Krishnakant Pal</Participantid><Country>United States</Country><Sfempname>Richard Harrison</Sfempname><City>Miami</City><Createdon>0000-00-00</Createdon></Empdetails></EmpdetailsSet>. There are buttons for 'Download', 'Download Payload', and 'Log Content'.

Figure 12: Successful XML to CSV Conversion Response

10. SFTP Upload

The generated CSV file is sent to the configured SFTP server.

The SFTP receiver is responsible for establishing the connection and uploading the processed file to the specified location.

The SFTP adapter is configured with the required server address, port, authentication method and credential.

The screenshot shows the 'SFTP' configuration window with the 'Target' tab selected. It is divided into 'FILE ACCESS PARAMETERS' and 'CONNECTION PARAMETERS'. Under 'FILE ACCESS PARAMETERS', 'Directory' is 'AEH', 'File Name' is 'KrishnakantPal.csv', and 'Append Timestamp' is unchecked. Under 'CONNECTION PARAMETERS', 'Address' is 'accenture191.files.com:22', 'Proxy Type' is 'Internet', 'Authentication' is 'User Name/Password', 'Credential Name' is 'SFTP_Invalid', 'Timeout (in ms)' is '10000', 'Maximum Reconnect Attempts' is '3', and 'Reconnect Delay (in ms)' is '1000'. There are also checkboxes for 'Automatically Disconnect' and 'Enable Support for Deprecated Algorithms', both of which are unchecked.

Figure 13: SFTP Receiver Configuration (Target Tab)

SFTP

General Target **Processing**

PROCESSING PARAMETERS

Change Directories Stepwise: ☒

Create Directories: ☒

Flatten File Names: ☐

Prevent Directory Traversal: ☒

Use Fast Exists Check: ☒

Handling for Existing Files: **Override** v

Use Temporary File: ☐

Figure 14: SFTP Receiver Configuration (Processing Tab)

SFTP File Verification

After successful processing, the generated CSV file was verified on the SFTP server. The file contains the employee data processed by the integration flow (KrishnakantPal.csv).

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	Mandt	Participantid	Country	Sfempname	City	Createdon													
2		500 Krishnakant Pal	United States	Richard Harrison	Miami	11/30/1999													
3																			

Figure 15: SFTP Receiver Configuration

11. Mail Notification

The integration flow uses a Mail receiver to send notifications after processing. Two notification scenarios are implemented:

Success Notification

A success email is sent when the integration completes successfully.

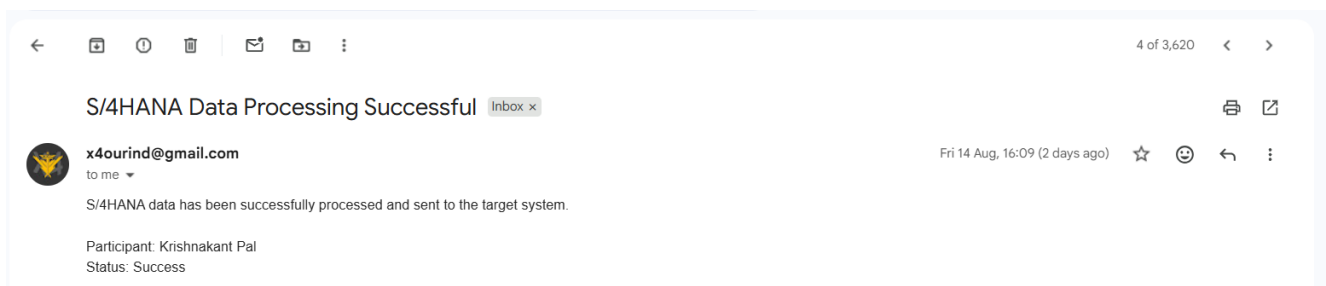


Figure 16: Successful Processing Email Notification

Failure Notification

If an exception occurs during processing, the Exception Subprocess prepares a failure notification and sends it through the Mail receiver.

12. Exception Handling

The integration flow uses an Exception Subprocess to handle runtime errors that may occur during processing.

When an error occurs, the exception flow is triggered automatically. The error information is processed using a Content Modifier and a failure notification is sent through the Mail receiver.

The exception integration flow is:

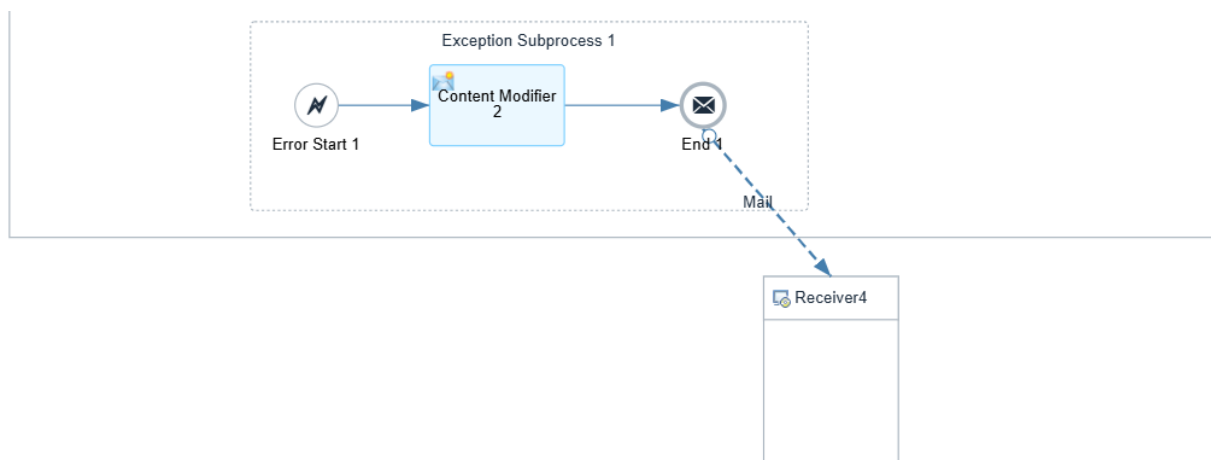


Figure 17: Exception Subprocess for Error Handling

Error Notification

When an exception occurs, the integration flow sends a failure email containing information about the failed processing.

Example:

S/4HANA Integration Failed.

An error occurred while processing the integration flow.

The failure notification helps identify that an issue occurred during integration processing.

13. Security and Credential Configuration

The integration uses configured security credentials for connecting to external systems.

Credentials are maintained separately from the integration flow configuration. The credential name is referenced by the respective adapter.

The following connections require authentication:

- SAP SuccessFactors
- SAP S/4HANA
- SFTP Server
- Mail Server

Note: Passwords, private keys and other sensitive authentication information are not included in this document.

14. Test Strategy

The integration flow was tested using both positive and negative test scenarios.

Positive testing was performed to verify that valid employee data is successfully processed from SuccessFactors to S/4HANA and that the generated CSV file is uploaded to the SFTP server.

Negative testing was performed by intentionally creating error conditions to verify the error handling and notification mechanism.

The main areas tested were:

- SuccessFactors data retrieval
- S/4HANA data processing
- Message Mapping
- XML to CSV conversion
- SFTP file upload
- Success email notification
- SFTP connection failure
- Duplicate Participant ID error
- Exception handling

15. Test Cases

Test Case Summary

Test ID	Test Scenario	Expected Result	Actual Result	Status
TC-01	Retrieve employee data from SuccessFactors	Employee data should be retrieved	Data retrieved successfully	PASS
TC-02	Send employee data to S/4HANA	S/4HANA should process the request	Request processed successfully	PASS
TC-03	Convert XML response to CSV	CSV file should be generated	CSV generated successfully	PASS
TC-04	Upload CSV to SFTP	File should be uploaded	File uploaded successfully	PASS
TC-05	Successful processing notification	Success email should be received	Email received	PASS
TC-06	Invalid SFTP credential/configuration	SFTP processing should fail	SFTP error generated	PASS
TC-07	Duplicate Participant ID	S/4HANA should reject duplicate data	Error received	PASS
TC-08	Exception handling	Failure notification should be generated	Failure email received	PASS

16. Test Execution and Evidence

Successful Integration Test

A successful execution of the integration flow was verified through the Cloud Integration Message Monitor.

The message processing status was shown as Completed.

The successful execution confirmed that the integration flow was able to process the data through the configured integration steps.

Status = Completed

Discover

Design

Monitor

Settings

Overview / Monitor Message Processing

Monitor Message Processing

Hide Filter Bar

Time: Past Hour

Status: All

Type: All

Package: All

Artifacts: Krishnakant_Pal_P20 x

or

ID: Message, Correlation or Application Message

Aug 14, 2026, 15:42:14 - Aug 14, 2026, 16:42:14

Messages (11)

Artifact Name	Status	Last Updated At	Artifact Type	Processing Time	Log Level	Sender	Receiver
Krishnakant_Pal_P2012786192	Completed	Aug 14, 2026, 16:40:58	Integration Flow	10 sec 894 ms	Info		
Krishnakant_Pal_P2012786192	Discarded	Aug 14, 2026, 16:40:48	Integration Flow	273 ms	Info		
Krishnakant_Pal_P2012786192	Discarded	Aug 14, 2026, 16:40:48	Integration Flow	277 ms	Info		
Krishnakant_Pal_P2012786192	Discarded	Aug 14, 2026, 16:40:47	Integration Flow	290 ms	Info		
Krishnakant_Pal_P2012786192	Discarded	Aug 14, 2026, 16:40:47	Integration Flow	280 ms	Info		
Krishnakant_Pal_P2012786192	Completed	Aug 14, 2026, 16:09:09	Integration Flow	14 sec 530 ms	Trace		
Krishnakant_Pal_P2012786192	Discarded	Aug 14, 2026, 16:08:59	Integration Flow	270 ms	Trace		
Krishnakant_Pal_P2012786192	Discarded	Aug 14, 2026, 16:08:55	Integration Flow	273 ms	Trace		
Krishnakant_Pal_P2012786192	Discarded	Aug 14, 2026, 16:08:55	Integration Flow	270 ms	Trace		
Krishnakant_Pal_P2012786192	Discarded	Aug 14, 2026, 16:08:55	Integration Flow	267 ms	Trace		
Krishnakant_Pal_P2012786192	Discarded	Aug 14, 2026, 16:08:54	Integration Flow	266 ms	Trace		

Figure 18: Successful Integration Flow Execution

SFTP File Path Verification

The generated CSV file was verified after successful processing.
The file was available on the configured SFTP server.

FILES

Search files, folders, and settings...

Ask AI

Onboarding

Files

Send & Receive

Partners

Automation & Flow

Integrations

Users & Groups

Encryption

AI

Clients & Protocols

AEH

Go up to parent folder

Files

History

Permissions

Settings

New

Name

Type

Last Modified At

Size

Actions

| KrishnakantPal.csv | CSV file | Aug 14, 2026, 6:39 AM | 123 B | |
| Displaying 1 item (1 file) | | | | |

Figure 19: CSV File Generated and Uploaded to SFTP

17. Negative Test – SFTP Failure

An SFTP failure scenario was tested by using an invalid SFTP credential/configuration.
During execution, the integration flow was unable to establish the required SFTP connection and generated an error.
The error was visible in the processing result and was also used to verify the exception handling mechanism.

S/4HANA Integration Failed

x4ourind@gmail.com

Fri 14 Aug, 16:40 (2 days ago)

S/4HANA Integration Failed.

An error occurred while processing the integration flow.

Error Message:
Failed to create Producer for endpoint: sap-sftp://accenture191.files.com/22/AEH?allowDeprecatedAlgorithms=false&authMethod=user&autoCreate=true&connectTimeout=10000&connectivityServiceProvider=%23connectivityServiceProvider&credential=SFTP_Invalid&disconnect=false&fastExistsCheck=true&fileExist=Override&flatten=false&jailStartingDirectory=false&keyStoreAccess=%23keyStoreAccess&knownHostsAccess=%23knownHostsAccess&maximumReconnectAttempts=3&passwordStorage=xxxxxx&pdUri=&proxyPort=8080&proxyProtocol=socks5&proxyType=internet&reconnectDelay=1000&soTimeout=300000&stepwise=true&tempFile=JHimaWxlOm5hbWV9LnRtcA%3D%3D&useTempFile=false. Reason: com.sap.it.nm.types.NodeManagerException: Error retrieving the artifact descriptor for SFTP_Invalid

Status: Failed

Figure 20: SFTP Connection Failure

18. Negative Test – Duplicate Participant ID

A duplicate Participant ID scenario was tested to verify the target system's validation and error handling.

When an already existing Participant ID was used, S/4HANA returned an error response indicating that the record could not be created because of the duplicate data.

The error was captured by the integration flow and processed through the Exception Subprocess.

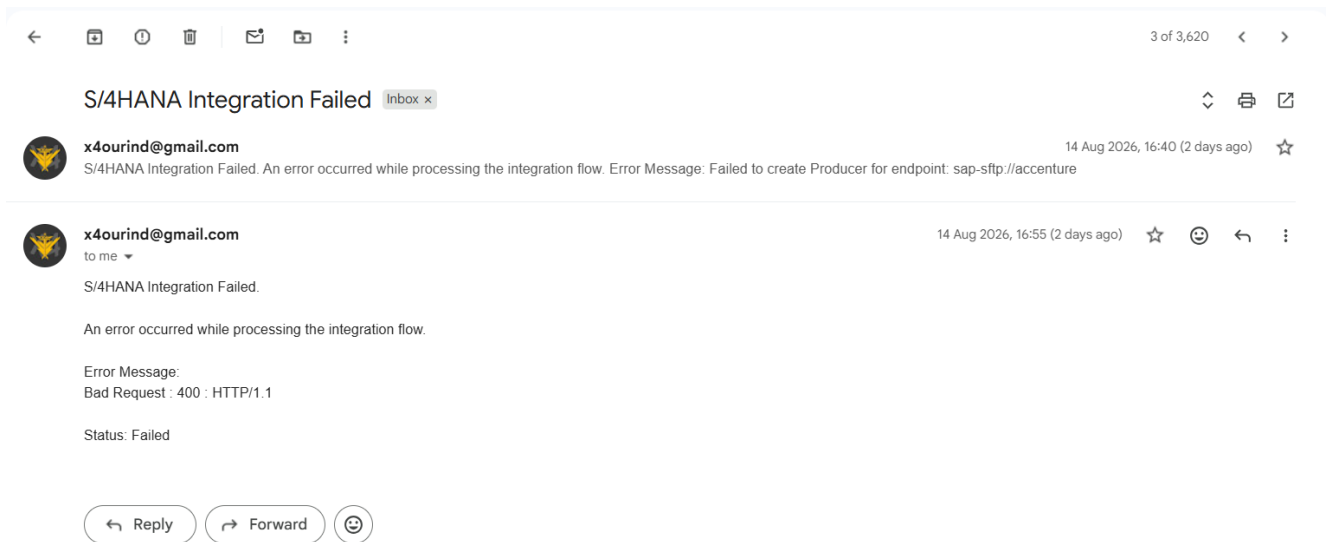


Figure 21: Duplicate Participant ID Error from S/4HANA

19. Failure Email Verification

The exception handling mechanism was verified by checking the failure notification received through email.

The email confirmed that the integration flow encountered an error during processing.

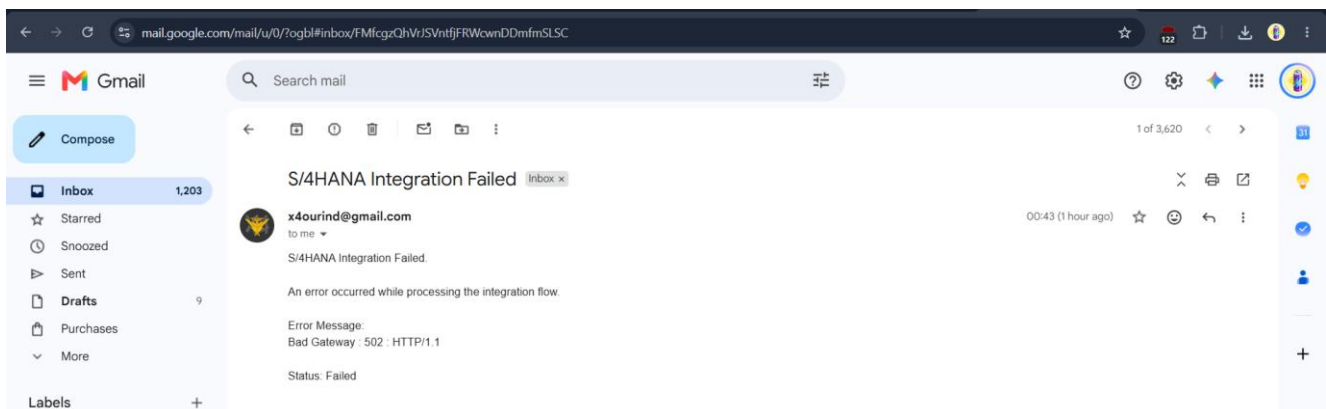


Figure 22: Failure Email Notification

20. Test Result Summary

All major integration scenarios were executed successfully.

The positive test cases verified the normal data processing flow, while the negative test cases verified the error handling and notification mechanism.

Category	Result
SuccessFactors Data Retrieval	PASS
S/4HANA Integration	PASS
Message Mapping	PASS
XML to CSV Conversion	PASS
SFTP Upload	PASS
Success Notification	PASS
SFTP Error Handling	PASS
Duplicate Participant ID Handling	PASS
Failure Notification	PASS

Overall Result

Integration Testing Status: **PASS**

21. Conclusion

The project successfully implements an integration solution using SAP Integration Suite – Cloud Integration.

The solution retrieves employee data from SAP SuccessFactors, transforms the data using Message Mapping, sends it to SAP S/4HANA, converts the processed data into CSV format, and uploads the file to an SFTP server.

The project also implements email notifications and an Exception Subprocess for handling runtime errors.

Both successful and negative test scenarios were executed to verify the reliability of the integration flow and its error-handling mechanism.

22. Appendix

Supporting screenshots and configuration evidence are included in the relevant sections of this document.